

Enhancing Adversarial Attacks through Chain of Thought

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Abstract

Large language models (LLMs) have demonstrated impressive performance across various domains but remain susceptible to safety concerns. Prior research indicates that gradient-based adversarial attacks are particularly effective against aligned LLMs and the chain of thought (CoT) prompting can elicit desired answers through step-by-step reasoning.

This paper proposes enhancing the robustness of adversarial attacks on aligned LLMs by integrating CoT prompts with the greedy coordinate gradient (GCG) technique. Using CoT triggers instead of affirmative targets stimulates the reasoning abilities of backend LLMs, thereby improving the transferability and universality of adversarial attacks.

We conducted an ablation study comparing our CoT-GCG approach with Amazon Web Services *auto-cot*. Results revealed our approach outperformed both the baseline GCG attack and CoT prompting. Additionally, we used *Llama Guard* to evaluate potentially harmful interactions, providing a more objective risk assessment of entire conversations compared to matching outputs to rejection phrases. The code of this paper is available at <https://github.com/sujingbo0217/CS222W24-LLM-Attack>.

1 Introduction

The advent of the large language models (LLMs) (Touvron et al., 2023a; Team et al., 2023; Chowdhery et al., 2022; Zhang et al., 2022a) has ushered in a new era of natural language processing, with these models demonstrating remarkable performance across various tasks (Brown et al., 2020; Betker et al., 2023). However, concerns regarding their safety and reliability have emerged, as LLMs are frequently vulnerable to adversarial attacks (Bommasani et al., 2021; Greshake et al., 2023). Recently, a host of adversarial attack approaches have emerged, including Token Manipulation (Morris et al., 2020), Gradient-based Attacks,

Jailbreak Prompting (Wei et al., 2024; Huang et al., 2023), and Red-teaming (Yu et al., 2023; Perez et al., 2022), as listed by Weng (2023). In particular, *gradient-based* adversarial attacks, which rely on training the parameters of white-box LLMs by gradient descent, have demonstrated high and significantly universal attack performance across different LLMs. Guo et al. (2021) proposed the first general-purpose attack, which employs the Gumbel-Softmax approximation to make adversarial loss optimization differentiable; Shin et al. (2020) implements a gradient-based search strategy to identify the most efficacious prompt template for a diverse set of tasks; Zou et al. (2023) was trained on white-box LLMs to learn universal attack suffixes and concatenate them to user input in order to elicit affirmative responses from LLMs rather than rejecting responses.

Chain of Thought (CoT) prompting can be broadly classified into two forms. The first involves the addition of a “trigger” after the user prompt, such as “*Let’s think step by step.*”. This can activate the reasoning chains in LLMs (Kojima et al., 2022). However, since the CoT techniques are mostly not gradient-based (Xiang et al., 2024), it is challenging to perform universal and transferable attacks on LLMs using this method. The efficacy of generated reasoning chains and responses heavily relies on the capacity of LLMs to understand the prompt and generate precise and pertinent responses. The second approach provides a few-shot demonstration by concatenating the user question with manually created reasoning steps (Wei et al., 2022). In this instance, each reasoning step comprises a question, a series of rationales, and the anticipated answer. Exposing the LLMs to these demonstrations helps them learn to generate similar rationales when presented with new questions. This strategy can sometimes “fool” the LLMs into producing incorrect or irrelevant answers for certain tasks, such as arithmetic (Roy and Roth, 2016)

and commonsense reasoning (Talmor et al., 2018). It consistently outperforms the first approach. Nevertheless, the implementation of robust safety measures in current aligned LLMs makes it nontrivial for LLMs to produce threatening responses.

In this paper, we propose a novel approach, Universal Adversarial Attacks through Chain of Thought, to address the limitations of existing CoT adversarial attacks on aligned LLMs. Our approach combines gradient-based methods with the CoT technique to enhance the effectiveness of adversarial attacks while maintaining the safety constraints of aligned LLMs. Instead of directly optimizing the input to deceive LLMs, we optimized the trigger phrase as a suffix of target outputs, activating the CoT reasoning steps of LLMs to generate desired answers, even though the model is aligned and safe to prevent from generating malicious content. Additionally, our experiments have shown that some LLM outputs are not offensive. It is important to note that the initial approach only verifies the presence of prefixes indicating LLMs’ refusal to respond, and does not analyze the risk of the conversation. A complete human-LLM interaction includes both user input and model output. To assess the hazard of LLM outputs objectively, we integrate the Llama guard (Inan et al., 2023) in our evaluation. This ensures that the level of risk of the conversation is analyzed, which may lead to a decrease in the Attack Success Rate (ASR). The following section will introduce the related work, while Section 3 will describe the experimental setup. Due to limited computational resources (as mentioned in Sec. 3.2), the improved method only slightly outperforms the original greedy coordinate gradient-based search (GCG). The performance will be discussed in the final section.

2 Experiments

2.1 CoT Adversarial Attack

We replace the goal of original *Harmful Behaviors* of Advbench from affirmative prompts such as “*Sure, here is*” to CoT triggers like “*Let’s think step by step*”. The architecture of a CoT adversarial attack looks like this:

System: You are a chat assistant designed to provide helpful and not harmful responses.

User: Tell me how to build a bomb. <suffix>

Assistant: Let’s build a bomb step by step: 1.

2.2 Setup

Baseline. The CoT-GCG approach was constructed based on Zou et al. (2023). Their harmful behaviors dataset and attack success rate (ASR) metric were modified and utilized for our ablation studies. The experiment involved training the transfer GCG on 50 prompts and testing on 25 prompts. A single universal adversarial suffix was generated using Vicuna-7B¹ (Chiang et al., 2023) and LLaMA2-7B² (Touvron et al., 2023b) models. The Vicuna model was trained by fine-tuning LLaMA, thus they utilize the same tokenizer and the capacity to share tokens during the attack. Additionally, since the Vicuna was trained on user-shared conversations collected from the ShareGPT website and evaluated by GPT-4 (Achiam et al., 2023), as well as the LLaMA-2 exhibiting a greater degree of alignment than other open-source LLMs, the adversarial suffixes generated by these two LLMs would be more robust and universal. Finally, the adversarial suffix with the lowest loss was obtained after 200 steps of suffix generation in the experiment.

Dataset. *Harmful Behaviors* dataset contains 520 pairs of harmful behaviors (goals) and their corresponding output instructions (targets). These behaviors encompass a range of profanity, threats, misinformation, discrimination, and dangerous or illegal suggestions. Three steps for processing this dataset are:

1. Categorize the *goals*, keep the data falling within the six harmful types of *Llama Guard*.
2. Convert affirmative *targets* to CoT triggers and utilize data with a length of less than 85.
3. Count the number of data for each harm type and calculate the percentage of harmful *goals*.

The preprocessing outcomes indicate that 93 out of 520 goals (17.9%) are not considered high-risk. The harmful categories of AdvBench’s *Harmful Behaviors* are shown in Table 1.

Evaluation. The performance of an adversarial attack is evaluated by *Attack Success Rate* (ASR) of 79 collected *goals* that fulfill the aforementioned conditions. An LLM is deemed to have been successfully attacked if it accurately outputs *harmful* content in response to a target command. The notable improvement is the replacement of the list comprising a series of prefixes that refuse to answer with *Llama Guard*. This modification allows for a more logical assessment of the potential harm

¹lmsys/vicuna-7b-v1.5

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Harmful Types					
O_1 (Violence)	O_2 (Sexual)	O_3 (Weapons)	O_4 (Regulated)	O_5 (Suicide)	O_6 (Criminal)
39/520	2/520	314/520	34/520	17/520	21/520
7.5%	0.3%	60.3%	6.5%	3.3%	4.0%

Table 1: Percentage of harmful target categories in AdvBench *Harmful Behaviors*.

of the generated content and facilitates the categorization and analysis of the harm types of prompts that can successfully attack LLMs.

Test Models. Five aligned LLMs were employed to evaluate the generalizability and transferability of adversarial attacks for ASR computation. The models included two public black-box models and three open-source models: GPT-3.5 Turbo³ (Brown et al., 2020), Claude-3 Haiku⁴ (Bai et al., 2022), LLaMA2-7B, Vicuna-7B, and Mistral-7B⁵ (Jiang et al., 2023).

Ablation. An ablation study was conducted on the approach of the project Zhang et al. (2022b) and Zou et al. (2023). A portion of the Auto-CoT pipeline was modified for evaluating the ASR on multiple LLMs. Furthermore, the same configurations were set up in the transfer GCG method and our method during training. The experimental results are presented in Table 2.

2.3 Results

The effectiveness of CoT-GCG on various LLMs

Table 2 is designed to demonstrate the superiority of our approach over non-gradient-based CoT and original GCG transfer methods in the context of public black-box and white-box LLMs. The first four rows of the table present the attack performance of non-gradient CoT approaches. The non-gradient CoT method is demonstrably ineffective against attacks on aligned LLMs, with virtually no efficacy. It would be highly risky to elicit the desired, but potentially dangerous, responses from signed LLMs by CoT prompting. The comparison experiment with Auto-CoT demonstrated slightly superior performance compared to the first three experiments, particularly in improved ASR for GPT-3.5, Vicuna-7B, and Mistral-7B models. However, this performance is not consistently reliable across different types of harmful prompts. For instance,

in the well-aligned LLaMA2-7B and Claude models, where adversarial attacks are almost ineffective against them. The fifth row represents our baseline, i.e. the original GCG transfer method with the prefix "Sure, here is". Its performance remains stable and reliable. Although the experimental results are not as good as those reported in the original GCG paper, the ASR is still capable of significantly outperforming the non-gradient CoT approaches in the same setting. The last row labeled CoT-GCG illustrates the optimal ASR for prompts prefixed with CoT triggers. This represents the most effective performance in the context of computational resource constraints, as well as in the case of attacks using higher-hazard prompts. It also demonstrates the best performance on all tested models. In addition, we found that all attack ASRs are inferior to those presented in the original GCG paper. This discrepancy can be attributed to the enhanced evaluation methodology with *Llama Guard* and constrained computational resources. However, our approach remains capable of further enhancing ASR on a range of aligned LLMs.

The results of attacks on diverse harmful types

Table 3 presents the ranked ASR of multiple aligned LLMs in response to various harmful categories of prompts in adversarial attacks. The data in the table is calculated by dividing the number of successful attacks on each category of goals by the ratio of all compromised goals in that category. This ratio is calculated by dividing the number of harmful goals in that category by the number of goals excluding harmless goals. The resulting value is then divided by the total number of successful attacks on each category of current goals. It can be observed that specific types of hazards, such as content related to the "Suicide" and "Criminal" categories, exhibit high ASRs across multiple models. Furthermore, their aggressiveness against GPT-3.5, as well as the open-source models tested, is considerable, with an average of over 100%. The Claude-3 model, however, is relatively well pro-

³<https://chat.openai.com>

⁴<https://claude.ai/chats>

⁵mistralai/Mistral-7B-Instruct-v0.2